

Measuring how Project Starline improves remote communication with behavioral science

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1 INTRODUCTION

Project Starline is a technology project that creates a remote communication experience that feels like you're together, even when you're thousands of miles apart. This perception of co-presence is created by representing users in 3D at natural scale, enabling eye contact, and providing spatially accurate audio. But to what extent do these technological innovations translate to meaningful, observable improvement in user value compared to traditional video conferencing?

Operationalizing the user value of a novel remote communication technology beyond self-report can be difficult because human communication relies partly on unconscious processes like nonverbal behavior. Indeed, some of our participants in early qualitative studies believed Project Starline provided benefits but had difficulty in fully articulating their experience. Thus, we conducted a number of studies using behavioral science (e.g., experimental design with behavioral outcomes). We found converging evidence that Project Starline outperforms traditional video conferencing in terms of conversation dynamics, video meeting fatigue, and attentiveness.

2 IMPROVED CONVERSATION DYNAMICS

Preliminary data suggest Project Starline improves conversation dynamics by increasing rates of turn-taking. We recruited pairs of participants who had never met each other to have unstructured conversations in both Project Starline and traditional video conferencing. We analyzed the audio from each conversation and found that Project Starline facilitated significantly more dynamic "back and forth" conversations compared to traditional video conferencing. Specifically, participants averaged about 2-3 more speaker hand-offs in Project Starline conversations compared to those in traditional video conferencing across a two minute subsample of their conversation. We also included self-report measures as a way to help interpret these behavioral findings. Participants rated their Starline conversations as significantly more natural ("smooth,"

"easy," "not awkward"), higher in quality, and easier to recognize when it was their turn to speak compared to conversations using traditional video conferencing.

In another study [Lawrence et al., 2021], participants had conversations with a confederate in both Project Starline and traditional video conferencing. We recorded these conversations to analyze select nonverbal behaviors [Hall et al., 2019]. In Project Starline, participants were more animated, using significantly more hand gestures (+43%), head nods (+26%), and eyebrow movements (+49%). Participants also reported a significantly better ability to perceive and convey nonverbal cues in Project Starline than in traditional video conferencing. Together with the turn-taking results, these data help explain why conversations in Project Starline may feel more "natural," a word commonly used by participants in our earlier qualitative studies.

3 REDUCED VIDEO MEETING FATIGUE

The causes of video meeting fatigue are complex [Riedl, 2022], but one possibility is that video communication is cognitively taxing because it becomes more difficult to convey and interpret nonverbal behavior. Considering previous findings that suggested Project Starline might improve nonverbal communication, we examined whether video meeting fatigue might also be improved (i.e., reduced) compared to traditional video conferencing.

Our study found preliminary evidence that Project Starline indeed reduces video meeting fatigue. Participants held 30-minute mock meetings in Project Starline and traditional video conferencing. Meeting content was standardized across participants using an exercise adapted from academic literature [Lafferty et al., 1974] that emulates key elements of a work meeting, such as brainstorming and persuasion. We then measured video meeting fatigue via the Zoom Exhaustion and Fatigue (ZEF) Scale [Fauville et al., 2021]. Additionally, we measured participants' reaction times on a complex cognitive task [Stoet, 2017] originally used in cognitive psychology. We repurposed this task as an exploratory proxy for video meeting fatigue based on the assumption that more fatigue would lead to slower reaction times. Participants reported significantly less video meeting fatigue on the ZEF Scale (-31%) and had faster reaction times (-12%) on the cognitive task after using Project Starline compared to traditional video conferencing.

4 INCREASED ATTENTIVENESS

In our earlier study on nonverbal behavior, we included an exploratory information-retention task. We asked participants to write as much as they could remember about each conversation (one in Project Starline, and one in traditional video conferencing). We found that participants wrote 28% more in this task after their conversation in Project Starline.

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To explore the concept of attentiveness further, we conducted a study in which participants wore eye-tracking glasses. This allowed us to calculate the percentage of time participants spent focusing on their conversation partner's face, an important source of social information in human interaction. Participants had a conversation with a confederate in Project Starline, traditional video conferencing, and in person. We found that participants spent a significantly higher proportion of time looking at their conversation partner's face in Project Starline (+14%) than they did in traditional video conferencing. In fact, visual attentiveness in Project Starline mirrored that of the in-person condition: participants spent roughly the same proportion of time focusing on their meeting partner's face in the Project Starline and in-person conditions.

5 CONCLUSION

Together, these findings offer a compelling case for Project Starline's value to users: improved conversation dynamics, reduced video meeting fatigue, and increased attentiveness. The use of behavioral

science to better understand how a new technology impacts users on dimensions they may find difficult to articulate (e.g., nonverbal behavior) uncovered these findings in ways that typical user research (e.g., surveys, interviews) could not, providing a compelling case study on how studying behavior can inform user value.

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